



Concurrent Aerobic and Strength Training for Body Composition and Health

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Introduction

Concurrent endurance and strength training (CT) has been used to enhance athletic performance in a variety of sports, but its use as a health-promoting strategy for the general population should not be overlooked. By combining endurance and strength exercise modes, CT stresses both the cardiovascular and the neuromuscular systems. Besides traditional adaptation in endurance parameters (e.g. maximal aerobic capacity, lactate threshold, exercise economy) and skeletal muscle function (maximal strength, muscle power, muscular endurance, etc.), CT can also influence body composition, as well as traditional and non-traditional cardiometabolic risk factors. This chapter will discuss the effects of CT on health-related outcomes, with an emphasis on body composition and cardiometabolic biomarkers.

Body Composition

Total Fat Mass and Percentage of Body Fat

Obesity has become a worldwide problem, and besides being a modifiable risk factor estimates from the World Health Organization are that obesity has increased dramatically in the past decades, affecting approximately 13% of the world's adult

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population in 2014, with nearly 39% of adults characterized as overweight [1]. Obese individuals are at greater risk for metabolic, orthopaedic and cardiovascular diseases (CVD), as well as some types of cancer [1, 2]. Reducing body fat is of particular relevance for those who are overweight and obese, and incorporating well-designed exercise training routines as part of their weight loss and management agenda can be an effective strategy [3]. Early investigations have employed skinfold thickness to estimate relative changes in body fat mass with CT [4, 5] and although this is a valid method, recent experiments have taken advantage of the more robust dual-energy X-ray absorptiometry (DXA) technique to track changes in body composition [6–8]. This section will focus on results from experiments employing DXA or imaging techniques for the assessment of total and regional fat distribution.

The effect of CT on total fat mass has been investigated in a variety of studies, with controversial results found in the literature. Donges et al. [6] reported reductions in total body fat mass of middle-aged and older men performing 3 months of endurance training (ET) or CT, while increases in muscle mass were only observed in those allocated to strength training (ST) or CT programmes. Similar benefits of CT in decreasing fat mass or percentage of body fat have been observed by others [8–11], and the superiority of CT over single-mode training routines (e.g. ST) to reduce total body fat mass has been a matter of debate, which may relate to the larger overall training volume performed by the CT group compared to ET or ST alone (i.e. participants engaged in CT performed the sum of each single-mode training routine) in some but not all experiments. Pooled data from 21 studies, however, indicate that CT brings about a similar reduction in body fat compared to ET or ST, but insufficient data precluded the analysis of performing different CT volumes in comparison to single modalities in terms of reducing body fat [12]. Furthermore, some studies have reported no alterations in body fat of individuals engaged in long-term CT [7, 9]. Instead of representing an inability of CT to ameliorate body composition, these divergent findings actually reflect differences in exercise and methodological protocols, as will be discussed below.

Several of these experiments were not designed to directly tease out the influence of exercise upon body fat mass and did not control for caloric intake, energy expenditure and potential compensatory behaviours that may increase or decrease one's energy balance. If participants' energy balance is not precisely controlled, any response (from decrease to increase) in body composition could be expected. Following thermodynamics, alterations in body mass are regulated by one's net energy balance. Loss of total body and fat mass is attributed to a prolonged negative energy balance [3, 13], with experimental evidence for this given by investigations in which energy intake, as well as expenditure, of participants is manipulated to introduce an energy deficit or surplus over a period of time. For example, Ross et al. [3] demonstrated that diet or exercise alone can bring about equivalent reductions in body fat, as long as a similar energy deficit is sustained.

That said, it seems evident that for any individual seeking exclusively for reducing his/her fat mass, one of the best options is to select exercise routines that generate the greatest total energy expenditure. Although not limited to ET (e.g. high-intensity interval training), increase in muscle aerobic capacity and mitochondrial content have been traditionally linked to chronic endurance exercise [14], and

ET may shift fuel utilization towards fat oxidation at rest and during submaximal exercise [15, 16]. Conversely, hypertrophic gains due to ST contribute to increase total energy expenditure, as fat-free mass is a major determinant of resting metabolic rate [17], although the absolute contribution of increasing skeletal muscle mass to total metabolic rate is relatively small [18]. As such, endurance exercise engaging a large muscle mass, such as walking/running and swimming, may be the most obvious choice to increase energy expenditure as it allows a relatively high metabolic demand to be sustained for prolonged periods, although similar reductions in fat mass would be expected from isocaloric strength exercise sessions. In a meta-analysis of CT studies, Wilson et al. [12] have estimated marginally superior decreases in body fat in response to strength training performed concurrently with running compared to cycling. Although running engages greater amounts of muscle mass, it may also impair hypertrophic gains to a greater extent in CT routines [12]. Hence, there is a trade-off between acutely increasing energy expenditure with running and improving fat-free mass, and thus resting metabolic rate, with cycling. Moreover, the same authors identified that intensity of the endurance exercise component was directly associated with decreases in body fat, which probably reflects a greater energy expenditure if exercise duration was matched between conditions.

It is worth noting, however, that there is always an upper limit of exercise volume that one is able to perform, and it is less likely (though possible) that ordinary individuals will engage in extensive single modality training routines (e.g. >1.5 h/session) without losing motivation, not to mention the increased risk of injury related to repetitive orthopaedic strain that may result from the same movement pattern in long duration rhythmic activities. As such, CT may be a strategic alternative to increase one's weekly energy expenditure by enabling the incorporation of additional amounts and more diverse exercises in a single training session (when CT is performed on the same day), or by increasing training frequency (when the endurance and strength exercise components are performed on alternate days).

Concurrent training can also be used to reduce one's percentage of body fat by directly reducing adipocyte mass, increasing fat-free mass or a combination of both. The percentage of body fat reflects the proportion of fat stores relative to total body mass, in a way that an increase in skeletal muscle mass can result in a reduction of the percentage of fat, irrespective of changes in lipid stores. Since long-term CT may bring about increases in muscle mass with concurrent decreases in fat mass (Fig. 19.1), this training mode may be considered an ideal alternative to reduce the percentage of body fat [11].

Abdominal Fat Depots

The site of body fat distribution plays a role in overall health, with excessive adiposity around the waistline (abdominal obesity) being a major risk factor for CVD and metabolic disorders [19, 20]. The negative influence of central obesity as a cardio-metabolic risk factor most likely derives from excessive accumulation of visceral fat (visceral obesity) than that in the subcutaneous compartment in the abdominal region [20], thus, even though decreasing total body fat content is a major goal for

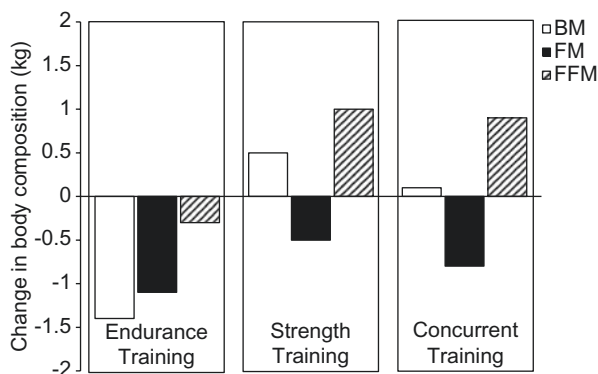


Fig. 19.1 Hypothetical model illustrating typical changes in selected body composition parameters to exercise training modalities. Endurance training engaging a large muscle mass holds a potential for large caloric expenditure, which may lead to a negative energy balance and reduce total body and fat mass (BM and FM, respectively). Strength training usually results in smaller or even an increase in total BM due to increases in fat-free mass (FFM), whereas concurrent training has the potential to ameliorate both FM and FFM. Ultimately, body composition changes will depend on a complex interaction between exercise modality and net energy balance

overweight and obese individuals, reducing or preventing excessive deposition of fat in the intra-abdominal compartment is considered of major relevance even for lean individuals. Visceral obesity is associated with waist circumference [21, 22], but it is more appropriately quantified through computerized axial tomography or magnetic resonance imaging as these techniques can be used to distinguish between fat in the subcutaneous and intra-abdominal compartment, and fair estimations can be obtained by DXA through android fat measurements [21].

Currently, a body of evidence indicates that exercise training can effectively reduce visceral fat depots. Results from a systematic review and meta-analysis reveal that endurance exercise appears to be effective in reducing abdominal fat, but the beneficial effect of ST is less evident [23]. Visceral fat reductions may be attained with CT but results are not as clear as with single-mode ET, most likely because of a reduction in the volume of the endurance exercise component during CT sessions in many studies. Nevertheless, randomized trials have confirmed the potential of CT to decrease central fat depots [6, 24] providing enough exercise volume is attained. Moreover, a recent meta-analysis of 15 trials comparing ET, ST and/or CT on metabolic and anthropometric parameters reported the latter to be effective in reducing waist circumference and suggests that CT is a more effective approach to prevent and treat obesity [25].

Recently, the distribution of strength and endurance exercise modes within a CT routine has received considerable attention from the scientific community (Fig. 19.2). Controversial results have been published regarding the influence of the CT loading order upon muscle mass and strength, with some research groups reporting greater adaptations when endurance exercise was preceded by strength exercises [26, 27], while others have found little or no exercise order effect upon neuromuscular parameters [7, 9, 28–31].

Loading Order	Day 1	Day 2	Day 3	Day 4
A				
B				
C				

Fig. 19.2 Example of concurrent endurance and strength training routines with different exercise loading orders. Same session concurrent training can be performed with the endurance component preceding strength exercises (A), or the reverse exercise sequence (B). Alternatively, each exercise mode can be performed in separate exercise sessions (C). The latter has been reported to optimize changes in android fat mass [9]

Different from the neuromuscular system, there appears to be consistent (yet limited) evidence regarding CT loading order and body composition. A series of Finnish studies has explored this topic and initially showed that the intrasession concurrent exercise sequence does not play a relevant role influencing changes in body fat, at least when an exercise routine with relatively low training volume is employed (i.e. 60–100 min of exercise per session, 2–2.5 sessions per week) [7]. A posterior study confirmed these findings and expanded the current knowledge by demonstrating that splitting the CT session so that the endurance and the strength exercise components were performed in different days resulted in superior fat mass reduction compared to volume-matched CT performed in the same session [9]. When the site of body mass change was taken into consideration, a greater reduction in android (central) fat mass was observed in both men and women performing the endurance and strength exercise on alternate days. These are intriguing results as each protocol was expected to have elicited similar exercise energy expenditure during the training sessions. One could speculate that participants performing CT in a single session could have exercised at lower relative intensity as prior endurance/high-intensity exercise can impair the performance of the subsequent strength component of CT [32], while maximal force capabilities may be less affected [32, 33]. Although excess post-exercise oxygen uptake (and rest energy expenditure) is affected by strength exercise intensity, Thornton et al. [34] estimated that the magnitude of the post-exercise energy expenditure difference between a high (e.g. 85% of 15RM) and low (45% of 15RM) intensity strength exercise session is minor. Thus, it is unlikely that intensity-related excess post-exercise oxygen uptake would account for the greater reduction in body fat when ST and ET are performed in

separate sessions. Alternatively, shorter but more frequent exercise bouts in the separate session CT group may have contributed to a greater total energy expenditure due to the cumulative effect of more frequent excess post-exercise oxygen consumption events, but this hypothesis remains speculative. Despite the uncovered underlying mechanisms, these are promising findings with practical implication for those interested in optimizing changes in body composition with exercise, but replication of these results in other populations more likely to benefit from reducing total and regional body fat (such as obese and patients with type 2 diabetes) is still necessary.

Cardiometabolic Health

Beyond increasing lean mass and reducing central fat content, CT bears the potential to improve additional risk factors for cardiovascular and metabolic diseases. Poor regulation of blood lipids, glucose metabolism, and elevated systemic blood pressure are among major modifiable cardiometabolic risk factors amenable through exercise. The next sections will focus on the interaction between CT and these health-related parameters.

Blood Lipids and Lipoproteins

Long-term exercise training is recommended to ameliorate blood lipid and lipoprotein concentrations. Results from cross-sectional studies have provided the first evidence that increased physical activity levels were associated with improved fasting lipid profile, with athletes displaying increased serum high-density lipoprotein cholesterol (HDL-c) values that may be as high as 50% greater than inactive control counterparts. Triglyceride (TG) concentrations fluctuate in the opposite direction, with those with higher fitness levels presenting lower blood TGs. There is less evidence to support consistent total cholesterol (TC) and low-density lipoprotein cholesterol (LDL-c) changes, and the benefits of being physically active is normally reduced or even disappear when results are adjusted for confounding factors [35]. Longitudinal experiments investigating exercise interventions generally agree with epidemiological data, but changes in lipids and lipoproteins are normally of smaller magnitude. Based on a meta-analysis of 25 studies, including 1404 apparently healthy participants, it seems safe to expect a modest (+2.5 mg/dL) though significant increase in HDL-c after long-term ET [36]. Conflicting results have been reported in randomized controlled trials regarding the ability of ST to alter blood lipids at rest, but an analysis of accumulated evidence suggested that ST may ameliorate basal TC, the ratio of TC/HDL-c, TG, LDL-c and non-HDL-c in adults [37]. A smaller number of studies have investigated the potential of CT to change one's fasting lipids and lipoproteins, but it is generally accepted that CT can improve parameters of dyslipidaemia in a variety of populations [10, 38, 39], which is not surprising considering the known effects of single ET and ST upon HDL-c and TC.

The protective effect of HDL-c against CVD likely reflects its role in reversal cholesterol transport, not to mention its antioxidant, anti-inflammatory and vasoactive functions. Simply put, the reversal cholesterol transport involves a mechanism by which stored cholesterol esters, for example in vascular wall macrophages, can interact with extracellular apolipoprotein A1 to form nascent HDL-c, which then continues to stock cholesterol to form mature HDL-c species [40]. Mature HDL-c can selectively deliver their cholesterol content to the liver for elimination through (1) a direct pathway, whereby efflux of cholesterol ester occurs directly from HDL-c to hepatocytes; or (2) by indirect pathways, including cholesterol-triglyceride exchange with other lipoproteins to produce cholesterol-rich LDL-c forms, which then can transfer cholesterol to hepatocytes [40].

It is reasonable to consider that the combination of endurance and strength exercise could exert an additive effect in improving fasting lipid profile. To compare the three different exercise modalities in blood lipids and lipoproteins adaptations in young overweight/obese but otherwise healthy individuals, Tseng et al. [39] conducted a 3-month training study in which participants were allocated to ET (5 day/week, ~60 min/day at 50–70% maximal heart rate), ST (5 day/week, ~60 min/session, multiple sets performed at 60–80% of one repetition maximum), CT (similar endurance training and strength training sessions performed on alternate days for a total of 5 day/week) or a control group. Although all exercise groups improved cardiometabolic parameters, the ET and CT groups presented greater increases in HDL-c compared to the ST group. Interestingly, participants in the CT group displayed greater decreases in fasting glucose and TG compared to the other two exercise modalities, even though the total training volume (estimated by exercise time in this study) of the CT group was not greater than the others. Similarly, Libardi et al. [41] reported that ST and CT benefited middle-aged men by reducing fasting TC, but only CT improved blood TG concentrations. These are important findings as they indicate that CT may be superior to ET and ST alone to improve lipid profile, but it is important to keep in mind that this premise does not always hold true as studies comparing the three modalities have also reported similar responses after an intervention period. In fact, meta-analyses by Schwingshackl and colleagues estimated that CT and ET were similar in improving blood lipids and lipoproteins in overweight and obese individuals [25], but CT was reported to be superior to ST in decreasing TG of type 2 diabetes mellitus patients [42].

Therefore, the current knowledge is that exercise training can lead to improvements in blood lipid and lipoprotein profile. Adding ST to an ET programme does not hamper these positive adaptations, and there is even evidence indicating that CT may be superior to each modality alone, although it is still uncertain whether this response reflects an additive interaction between adaptations from different training modalities or greater exercise volume with CT.

Glycaemic Control

Healthy individuals with preserved glycaemic control display fasting circulating glucose concentrations below 100 mg/dL, and glycated haemoglobin (HbA1c)

under 5.7%. Chronic impairments in normal glucose regulation, such as diabetes mellitus, place individuals at an increased risk of mortality and long-term complications as neuropathy, nephropathy, CVD and amputations. Lifestyle modifications including the adherence to a healthy diet and increasing physical activity levels are considered major non-pharmacological strategies in the prevention and treatment of type 2 diabetes mellitus and the metabolic syndrome [43–45]. Supervised exercise training has been shown to be particularly relevant to improve one's glycaemic regulation by decreasing fasting glucose and the absolute percentage of HbA1c in type 2 diabetic patients [46]. The effectiveness of ET to improve glucose metabolism in these patients has been documented for decades [47], and later evidence led to the recognition that supervised ST as well as the combination of endurance and strength exercises are effective alternatives in improving patient's glycaemic control [46].

As CT combines the benefits of ET and ST, it has the potential to optimize metabolic responses in type 2 diabetics. In a large trial comparing the three exercise modalities upon long-term changes in HbA1c, CT brought about superior improvements in glycaemic control compared to single endurance or strength exercise groups [48]. Furthermore, those with a lower initial fraction of HbA1c only benefited from CT. Although the referred work indicates superiority of CT, it is important to highlight that patients enrolled in the combined training group completed greater weekly exercise volumes compared to those allocated to single modality groups. Hence, it is difficult to determine whether the observed effects resulted from a superiority of CT modality or from a greater total work performed.

Attempting to answer the abovementioned question, Church and colleagues [49] conducted a 9-month exercise trial trying to match the total training volume between exercise programmes, so that the CT group expended less time engaged in the endurance and strength activities compared to single ET and ST groups. Only patients engaged in CT attained reductions in HbA1c compared to a non-exercise control group. A posterior study corroborated these findings by reporting greater improvements in fasting glucose concentration in healthy males after CT compared to ET or ST, even with the endurance and strength components reduced by half in the CT group [39]. These results indicate a potential superior effect of CT in glucose metabolism that is irrespective of exercise volume. Nevertheless, a meta-regression of randomized controlled trials identified CT volume, particularly of the strength component, to be associated with improvements in HbA1c of type 2 diabetes patients [50], underscoring the importance to consider the interaction between exercise volume and mode in chronic adaptations for glucose regulation.

Exercise intensity was initially identified as a predictor of improvements in HbA1c in endurance exercise trials [47], but not in a more recent meta-regression including CT trials [50]. Inconsistent results about the putative benefit of higher intensity CT programmes upon glucose regulation-related outcomes are found in the literature, probably because of the interaction between training intensity and volume which results in greater total work performed by a high-intensity compared to low-intensity groups if the same exercise duration were used. A subgroup analysis of the Italian Diabetes Exercise Study helped to shed some light upon this topic, as researchers compared the outcomes of type 2 diabetics performing energy

expenditure-matched CT at low vs. moderate-to-high intensity. After 12 months of exercise intervention, patients exercising at higher intensity displayed a greater reduction of -0.5% in Hb1Ac compared to -0.33% in the lower intensity CT subgroup. Although an intensity dose-response could be identified, the additional benefit of higher intensity CT was deemed of little clinical relevance by the researchers [38], but we stress that a $\sim 0.15\%$ difference in Hb1Ac should not be overlooked.

The benefit of CT in glycaemic control of apparently healthy individuals is less clear. There seems to be mixed results in the literature comparing CT to single exercise modalities, with greater improvements in fasting glucose concentrations of young males reported to occur after CT compared to ET or ST [39], while others found potentially greater benefits of ET in glucose disposal during an oral glucose tolerance test in men [6], or even superior effects of ST and CT in reducing fasting glucose concentrations of older women compared to ET [8]. In some situations, exercise training may not improve fasting glucose concentrations after either modality, but prevent an impairment in glycaemic control over time [41], which probably reflects a narrower range for physiological adaptation in middle-aged otherwise healthy individuals compared to patients with established metabolic disorders.

Taken together, CT seems to be at least as effective as ET and ST to improve glycaemic control (or prevent its decline) in healthy individuals. There is evidence that CT may be superior to either ST or ET alone in improving glucose-related outcomes in those with type 2 diabetes mellitus, but it is currently difficult to ascertain whether this adaptation represents a superior effect of CT per se, or simply reflects the potential greater total amount of exercise performed by concurrent trained individuals. Clearly, studies designed to directly compare different CT volumes are still necessary for more definitive conclusions to be taken.

Blood Pressure and Vascular Adaptations

High blood pressure is a major cardiovascular risk factor, associated with alterations in the structure and function of the vasculature. Traditionally, systemic hypertension has been defined as (sustained) rest office-based systolic and/or diastolic brachial artery blood pressure $\geq 140/90$ mmHg, respectively, but in 2017 the American College of Cardiology and the American Heart Association published new guidelines with more aggressive blood pressure thresholds for the US population, with hypertension characterized as systolic blood pressure ≥ 130 mmHg and/or diastolic blood pressure ≥ 80 mmHg [51]. The hypertensive population is more prone to adverse events like stroke, heart failure, peripheral artery disease and target organ damage [52]. Furthermore, increased haemodynamic forces can damage the vascular wall leading to endothelial dysfunction and result in further vascular health impairment [52]. A correlation between the increase in blood pressure and cardiovascular risk is observed even in the normotensive range, which stresses the need of well-suited approaches to prevent or delay progressive elevations in blood pressure [52].

Endurance exercise has been classically recommended as an essential part of the antihypertensive treatment and it seems safe to expect hypertensive patients to

benefit with reductions of 5–7 mmHg in resting blood pressure with long-term ET [53]. Contrary to earlier beliefs, the practice of dynamic ST does not promote chronic increases in blood pressure and can, thus, be used as an adjuvant blood pressure-lowering strategy with effects comparable to ET [53]. The combination of both modalities is well accepted and recommended for prevention and treatment of hypertension. A meta-analysis of 68 trials investigating the influence of CT on blood pressure estimated an overall reduction of -3.2 mmHg in systolic blood pressure, and -2.5 mmHg in diastolic blood pressure compared to non-exercise control groups when all participants were taken into consideration [54]. Importantly, the long-term antihypertensive effects of CT increased with baseline blood pressure levels, stacking up at -5.3 mmHg for systolic blood pressure, and -5.6 mmHg for diastolic blood pressure in hypertensive individuals, with decreases of almost 9 mmHg reported in the trials ranked as high methodological quality. These results suggest that those in greater need of blood pressure reductions will be the ones to benefit the most with training.

The mechanisms by which exercise brings about a chronic reduction in blood pressure must relate to alterations in cardiac output and/or peripheral vascular resistance [52]. Since resting cardiac output is affected minimally or not at all by exercise training [55], changes in vascular resistance may mediate most of the benefits of exercise. The long-lasting antihypertensive benefit of exercise results from a complex interaction of improved vascular responsiveness, neurohumoral adjustments and/or structural vascular adaptation [52, 55]. Several of such mechanisms are still being investigated, but particularly functional and structural vascular adjustments to exercise training have been well characterized. The vascular endothelium plays a pivotal role control in vascular health and haemostasis. Endothelial cells are prone to damage induced by chronic high blood pressure and dyslipidaemia, with endothelial dysfunction being related to myocardial infarction, hypertension, and even ageing without clinical evidence of CVD [56, 57]. The consequences of endothelial dysfunction are at least twofold: (1) an atherosclerotic-prone vascular bed; with (2) impaired vasodilatory capacity, which may contribute to development and progression of hypertension.

Impaired vascular function can be improved by either ET or ST alone [56, 58]. Acute dynamic exercise exerts direct stimulation on blood vessels by increasing vascular anterograde shear stress, which acts as a potent stimulus to trigger endothelial adaptations. Transitory increases in anti-atherogenic shear stress may occur even in non-exercising vascular beds, providing prolonged large muscle mass dynamic exercise is performed. Moreover, changes in systemic factors including an exercise-induced increase in circulating myokines (such as interleukin-6), as well as chronic reductions in pro-inflammatory agents (e.g. C-reactive protein, tumour necrosis factor α) and improvement in insulin signalling in endothelial cells all converge to exert a positive effect on the vasculature [59]. Besides acute haemodynamic adjustments, CT has the potential to ameliorate lipid and low-grade chronic inflammation markers in selected populations [6, 25] and can be considered an effective lifestyle strategy to increase endothelial function. Accordingly, individuals with risk factors for vascular dysfunction or established CVD generally display increased

endothelial-dependent vasodilation in conduit and resistance vascular beds after a short period of CT [60, 61]. Interestingly, there is evidence that the within-session CT loading order may influence vascular adaptations to exercise. Starting the CT training sessions by the strength component (i.e. strength + endurance exercise sequence) has been reported to lead to improvements in the endothelial function of young females and males, whereas no improvements in vascular function were observed with the loading order (endurance + strength) [62]. Although the precise underlying mechanisms remain elusive, it has been speculated that a potential attenuation of resistance exercise-associated increase in circulating growth hormone by prior endurance exercise [63] may have hindered endothelial adaptation in the endurance + strength training group, since growth hormone infusion has been reported to stimulate nitric oxide production (as estimated by nitrate balance) and endothelial-dependent vasodilation [64].

Although it remains debatable whether arterial stiffness contributes to or results from hypertension, it is established that augmented aortic stiffness is related to cardiovascular risk [65]. Deposition of collagen fibres and reduction of elastin content in the blood vessels eventually results in stiffer arteries with reduced capacity to accommodate pulsatile intravascular tension. High-intensity strength training is generally linked to increased arterial stiffness in young healthy adults [66] probably due to a compensatory adaptation to the large increases in blood pressure during exercise manoeuvres. Increased vascular stiffness in young individuals to ST is unlikely to represent a pathological adjustment, but alternatives to prevent this vascular remodelling process include the manipulation of ST intensity, as low to moderate intensity ST does not seem to impair central arterial compliance in young individuals [67], or selective addition of an endurance component to a ST routine [62, 68]. As reported by Kawano et al. [68], CT appears to be effective to prevent chronic reductions in arterial compliance, typically associated with high-intensity ST. Similar to vascular reactivity, Okamoto et al. [62] also observed that reductions in brachial-ankle pulse wave velocity (a surrogate marker of central arterial stiffness) after 8 weeks of CT was only evident in participants who followed the strength + endurance exercise sequence, suggesting that the CT loading order may exert some influence upon vascular function and remodelling. However, the underlying mechanisms leading to superior vascular adaptations with the strength + endurance exercise sequence reported by Okamoto and colleagues remain largely unknown.

Summary

Different from what may happen with maximal strength development, the combination of strength and endurance exercise training does not impair health-related adaptations. CT is effective to improve body composition, blood pressure, and blood lipid/lipoprotein profile, with some evidence that this exercise modality may surpass single endurance and strength exercise training for enhancing glucose regulation. Maintenance and even improvement in vascular health can also be attained

through concurrent exercise, and the endurance component of CT may be effective to prevent strength training-induced increases in arterial stiffness. In a pragmatic perspective, for reducing body fat the intensity of the CT endurance component may be a relevant variable, with short duration (20–30 min) endurance exercise performed at a higher intensity as a potential choice to reduce body fat while limiting a negative interference effect of CT on hypertrophic gains [12]. Those seeking for greater skeletal muscle and strength development may opt for combining strength and cycling training, whereas running as endurance modality may be slightly favourable to reduce body fat [12]. Even though CT volume and intensity can play a relevant role in improving glycaemic control and additional cardiometabolic risk factors in selected populations, there is evidence that CT per se has a superior effect compared to ST or ET alone. As such, a CT routine combines the benefits from each single exercise modality and may be considered an optimal training alternative for health improvement and maintenance.

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